Database development with PL/SQL INSY 8311

ADVENTIST UNIVERSITY OF CENTRAL AFRICA

Instructor:

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ORACLE PL/SQL

6h00 pm – 9h50 pm

- Monday A -G207
- Tuesday B-G204
- Wednesday E-G207
- Thursday F-G307

February 2025

Database development with PL/SQL

Reference reading

SQL*Plus

- [4 PL/SQL Control Statements]
- [Oracle PL/SQL Insert, Update, Delete & Select Into [Example]]
- YouTube Tutorial: [Oracle CDBs and PDBs Explained (With Connections)]
- [Oracle 12c: Applying PSU 12.1.0.1.1 with Multitenant DB & unplug/plug]
- [Multitenant : Overview of Container Databases (CDB) and Pluggable Databases (PDB)]
- [1 Introduction to the Multitenant Architecture]
- [Getting started with Oracle Database 12c Multitenant Architecture]

Course Objectives: CDB, PDB, and Oracle Enterprise Manager

Understand Container Database (CDB) and Pluggable Database (PDB) architecture and their role in Oracle multi-tenancy.

Create, configure, and manage CDBs and PDBs using SQL*Plus and Oracle tools.

Perform SQL operations within PDBs, including creating schema objects.

Administer databases using Oracle Enterprise Manager (OEM) for monitoring and management.

Monitor and optimize database performance using OEM tools.

Automate routine tasks and schedule jobs via OEM.

Manage users and security in CDB and PDB environments.

Execute backup and recovery operations at both CDB and PDB levels.

Troubleshoot common database issues using OEM diagnostic tools.

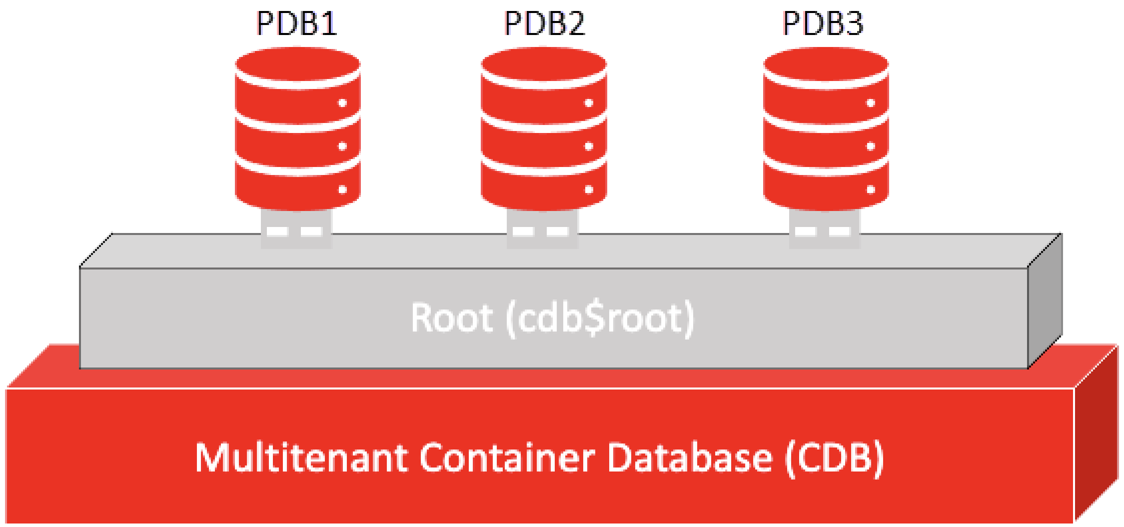
Oracle Databases CDBs and PDBs

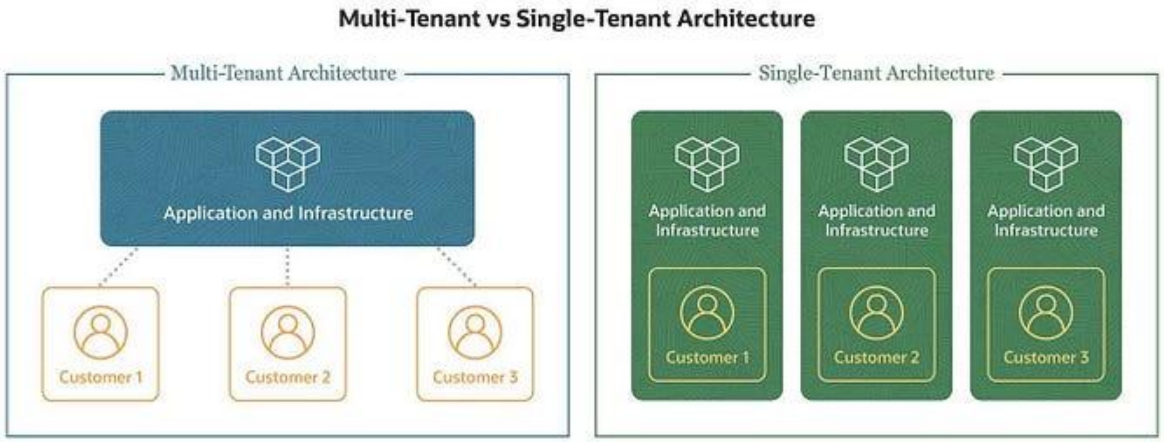
SQL*Plus

In Oracle databases, **CDBs (Container Databases) and PDBs (Pluggable Databases)** are part of Oracle's Multitenant Architecture, introduced in Oracle Database 12c. This architecture allows for better resource management, scalability, and easier database consolidation.

Multitenant Container Database Architecture

Oracle's **Multitenant Architecture** enables efficient management of multiple databases through a single **Container Database (CDB)**.





Source Image: http://medium.com/@davidadjirackor/oracle-single-tenant-vs-multi-tenant-architecture-explained-f83f2fc6e2b1

Oracle Single-Tenant (Non-CDB) vs Oracle Multi-tenant (CDB & PDB)

This is the traditional architecture where a single Oracle Database contains all components (data dictionary, user data, and system metadata) in one structure.

Features:
- Each database instance is independent.
- Requires separate system resources (memory, processes).
- Administrative overhead increases with multiple databases.
- Upgrades, patching, and backups must be done individually.
- Every database has its own set of background processes and system metadata

Pros:
✓ **Simplicity** – Each database is managed separately.
✓ **No additional licensing required.**
✓ **Works well** for smaller applications with low complexity.

Cons:
❌ **Resource-intensive** – Each database consumes separate CPU, memory, an disk resources.
❌ **Difficult to manage at scale** – Administering multiple databases is time-consuming.
❌ **Higher maintenance costs** due to separate backup, security, and upgrade efforts.

Oracle Multitenant allows multiple **Pluggable Databases (PDBs)** to share a single **Container Database (CDB)**. This enables database consolidation while maintaining separation between individual PDBs.

Features:
- A single CDB can host multiple PDBs.
- Shared memory and background processes, improving efficiency.
- PDBs can be unplugged and moved between CDBs.
- Centralized management with common users, roles, and privileges.
- Patch, upgrade, and backup at the **CDB level**, reducing operational overhead.

Pros:
✓ **Resource efficiency** – PDBs share CPU, memory, and processes, optimizing resource usage.
✓ **Easier management** – One container database to manage multiple pluggable databases.
✓ **Fast provisioning** – Quickly create, clone, or unplug PDBs.
✓ **Better security & isolation** – Each PDB remains logically separate while benefiting from shared infrastructure.
✓ **Simplifies upgrades & patches** – Apply changes to the CDB, affecting all PDBs at once.

Cons:
❌ **Requires Oracle Multitenant license** (except for one PDB in SE2/EE).
❌ **More complex setup** than single-tenant.
❌ **CDB failure impacts all PDBs** unless configured for high availability.

SQL*Plus

When to Choose Each? | Key Differences

Choose Single-Tenant (Non-CDB) if:

- You have a small number of databases.
- You prefer simpler database management.
- You don't need Oracle Multitenant features.

Choose Multitenant (CDB/PDB) if:

- You need to consolidate many databases efficiently.
- You want easier patching, cloning, and management.
- You plan to scale up database instances.
- You are using Oracle Cloud or a large enterprise system.

Feature	Single-Tenant (Non-CDB)	Multitenant (CDB/PDB)
Architecture	One database with its own resources	One CDB with multiple PDBs sharing resources
Resource Usage	Higher (each DB uses its own resources)	Lower (shared memory, processes, storage)
Management	Each database is managed separately	Centralized management for all PDBs
Scalability	Limited	High (easily add/remove PDBs)
Backup & Recovery	Per database	Can be done at CDB level for all PDBs
License Requirement	No additional license required	Requires Multitenant option (for more than one PDB)
Security Isolation	Fully separate instances	Logical separation between PDBs within a CDB
Upgrade/Patch	Done per database	Done once for all PDBs within a CDB

SQL*Plus

Multitenant, Structure of CDBs and their PDBs.

1.Oracle Instance (SGA/Processes):

1. This refers to the Oracle instance, which includes the System Global Area (SGA) and background processes that manage database operations.

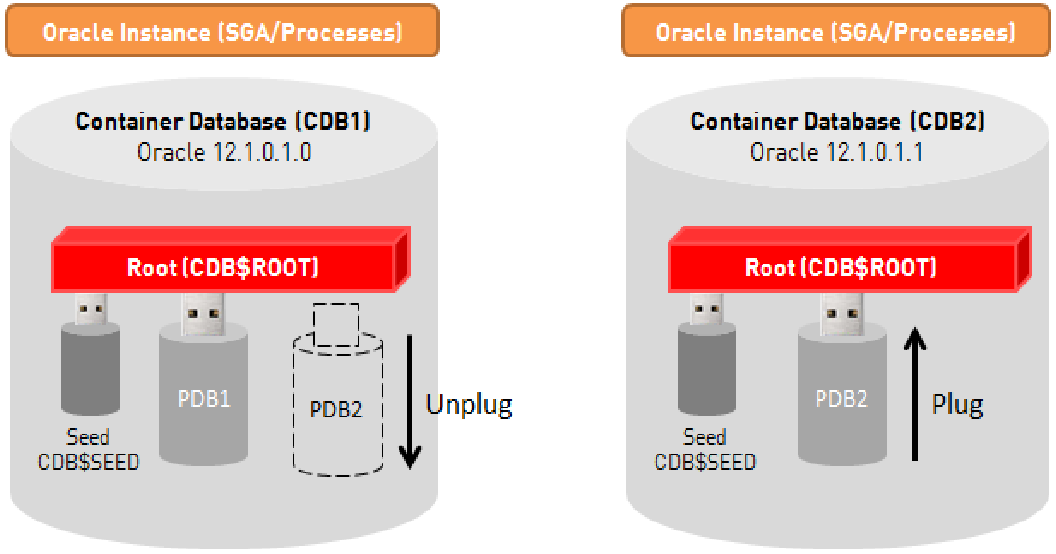
2.Container Database (CDB1):

1. **Oracle 12.1.0.1.0**: This indicates the version of the Oracle database.
2. **Root (CDB$ROOT)**: The root container, which is the main container in a CDB. It contains the system metadata and common users.
3. **Seed (CDB$SEED)**: A template PDB used to create new PDBs.
4. **PDB1 and PDB2**: These are Pluggable Databases within the CDB. PDBs are separate databases that can be managed independently but share the same Oracle instance.
5. **Unplug**: This suggests that a PDB can be unplugged from the CDB, meaning it can be detached and moved to another CDB.

3.Container Database (CDB2):

1. **Oracle 12.1.0.1.1**: A different version of the Oracle database.
2. **Root (CDB$ROOT)**: Similar to CDB1, this is the root container for CDB2.
3. **Seed (CDB$SEED)**: The template PDB for CDB2.
4. **PDB2**: A Pluggable Database within CDB2.
5. **Plug**: This indicates that a PDB can be plugged into the CDB, meaning it can be attached and used within this CDB.

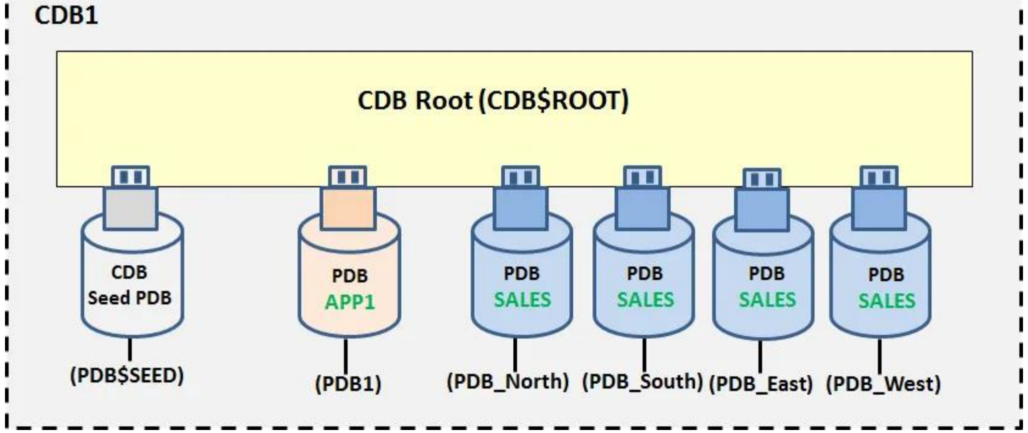
SQL*Plus



Source Image: https://www.dbi-services.com/blog/oracle-12c-applying-psu-121011-with-multitenant-db-and-unplugplug/

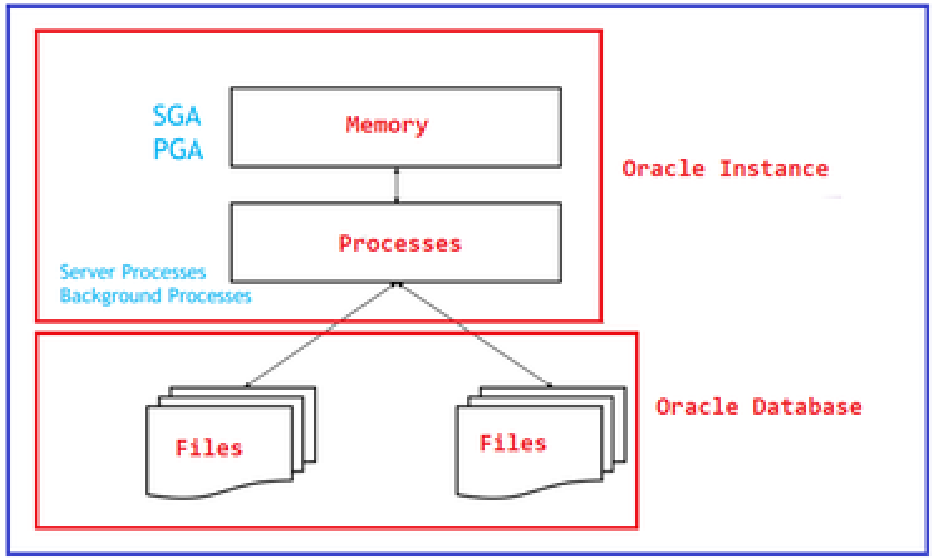
Key Differences: CDB, CDB$ROOT, PDB and PDB$SEED

Feature	CDB (Container Database)	CDB$ROOT (Root of CDB)	PDB (Pluggable Database)	PDB$SEED (PDB Seed)
Type	Top-level container for multiple PDBs	Root container for system-level data	A pluggable, fully functional database	Template for creating new PDBs
Purpose	Houses multiple PDBs and system metadata	Stores common system data and schemas	Holds user data, schemas, and apps	Used to create new PDBs
User Data	No user data	No user data	Stores user data	No user data
Modifiable	Read-write (but for managing PDBs)	Read-write (system-level management only)	Read-write	Read-only (cannot be modified)
Creation of New PDBs	Manages and contains PDBs	Contains only system schemas and data	Created from CDB$SEED or other PDBs	Used to clone and create new PDBs
Contained Objects	Contains the entire multitenant structure	Contains system data for all PDBs	Contains user-defined schemas, tables, etc.	Contains templates for new PDBs



Source Image: https://blog.toadworld.com/2017/06/26/oracle-multi-tenant-application-containers-part-i

How an Instance Works



Source Image: https://dotnettutorials.net/lesson/oracle-database-vs-database-instance/

SQL*Plus

Memory Structures:

System Global Area (SGA): This is a shared memory area that contains data and control information for an Oracle instance. The SGA is used by all background processes and user processes.

1. **Database Buffer Cache**: Stores data that is read from disk for faster access.
2. **Shared Pool**: Caches SQL queries, PL/SQL programs, and data dictionary information.
3. **Redo Log Buffer**: Stores redo log entries that are later written to disk.
4. **Java Pool**: Used for memory related to Java objects (if using Oracle Java).

Program Global Area (PGA): Memory used by an individual Oracle session. It stores session-specific data such as sort areas and session variables. Each session gets its own PGA.

Background Processes:

1. **Database Writer (DBWn)**: Responsible for writing modified data from the buffer cache to the data files.
2. **Log Writer (LGWR)**: Writes redo log entries from the redo log buffer to the online redo log files to ensure data is recoverable.
3. **Checkpoint (CKPT)**: Ensures that the data files are synchronized with the control files and that data is written to disk at appropriate times.
4. **System Monitor (SMON)**: Performs recovery after a system crash and cleans up temporary segments.
5. **Process Monitor (PMON)**: Cleans up after failed sessions and releases resources.
6. **Archiver (ARCn)**: If archiving is enabled, this process copies the filled redo log files to archive storage for backup and recovery.

User Processes:

- Each user session creates a user process that communicates with the Oracle instance to execute SQL queries. User processes do not interact directly with data files; they go through the instance's memory and background processes.

The Role of the Oracle Instance:

1. **Query Processing**: When a user submits a SQL query, the instance parses, optimizes, and executes it by reading data from the database or memory.
2. **Transaction Management**: Oracle ensures that transactions are consistent, durable, and isolated (ACID properties).
3. **Recovery**: The instance ensures that all transactions are properly logged and can be rolled back or recovered if the system crashes.

Display all PDBS

```
Command Prompt - sqlplus s

Microsoft Windows [Version 10.0.22631.4037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Eric>sqlplus sys as SYSDBA

SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 14:36:16 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle.  All rights reserved.

Enter password:

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> show user;
USER is "SYS"
SQL>
SQL> SELECT instance_name FROM v$instance;

INSTANCE_NAME
----------------
oracle21cdb

SQL> show pdbs;

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 ORACLE21DBCPDB                 READ WRITE NO
SQL> |
```

SQL*Plus

- To connect to an Oracle database using SQL*Plus as the SYS user with SYSDBA privileges.

sqlplus sys as SYSDBA
Or
sqlplus sys/admin@localhost:1521/ORACLE21CDB AS SYSDBA

- Oracle instance that is currently running, which is usually the same as the **SID** (System Identifier) of the database.

SQL> SHOW PDBS;

- This PDB helps streamline the process of managing multiple PDBs within a container database (CDB).

- This is the PDB created during the installation of Oracle 21c; for others, you might have XE.

Show path of PDBs

```
SQL> show pdbs;

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 ORACLE21DBCPDB                 READ WRITE NO
SQL> SELECT CON_ID, TABLESPACE_NAME, FILE_NAME
  2  FROM CDB_DATA_FILES
  3  WHERE CON_ID = 3;

    CON_ID TABLESPACE_NAME
---------- ------------------------------
FILE_NAME
--------------------------------------------------------------------------------
         3 SYSTEM
E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\SYSTEM01.DBF

         3 SYSAUX
E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\SYSAUX01.DBF

         3 UNDOTBS1
E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\UNDOTBS01.DBF


    CON_ID TABLESPACE_NAME
---------- ------------------------------
FILE_NAME
--------------------------------------------------------------------------------
         3 USERS
E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\USERS01.DBF


SQL>
```

- This query retrieves the tablespace names and file names of data files for the Pluggable Database (PDB) with CON_ID = 3 from the CDB_DATA_FILES view in an Oracle multitenant environment. - **Location of XE PDB1**

SQL> SELECT CON_ID, TABLESPACE_NAME, FILE_NAME FROM CDB_DATA_FILES WHERE CON_ID = 3;

- This path is commonly used for organizing database files, making it easier to manage multiple PDBs within a CDB.

Create PDB using PDBSEED

Command Prompt - sqlplus s

```
SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 17:30:49 2024
Version 21.3.0.0.0

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Enter password:

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> CREATE PLUGGABLE DATABASE PDB2
  2  ADMIN USER pdbadmin IDENTIFIED BY admin
  3  FILE_NAME_CONVERT = ('E:\oracle21cdata\ORACLE21CDB\pdbseed\',
  4                       'E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\PDB2\');

Pluggable database created.

SQL>
SQL>
SQL> SELECT pdb_name, status FROM cdb_pdbs;

PDB_NAME
--------------------------------------------------------------------------------
STATUS
----------
ORACLE21DBCPDB
NORMAL

PDB$SEED
NORMAL

PDB2
NEW
```

- This creates a new pluggable database by copying the structure of pdbseed and setting up a new admin user.

SQL> CREATE PLUGGABLE DATABASE PDB2
ADMIN USER pdbadmin IDENTIFIED BY admin
FILE_NAME_CONVERT =
('E:\oracle21cdata\ORACLE21CDB\pdbseed\',

'E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\PDB2\');

- It retrieves the name and status of all Pluggable Databases (PDBs) from the CDB_PDBS view in an Oracle multitenant environment.
- **PDB2 has status new because it has not yet opened**

SQL> SELECT pdb_name, status from cdb_pdbs

PDB Status

```
SQL> select name, open_mode from v$pdbs;

NAME
--------------------------------------------------------------------------------
OPEN_MODE
----------
PDB$SEED
READ ONLY

ORACLE21DBCPDB
READ WRITE

PDB2
MOUNTED


SQL> SELECT NAME, CON_ID FROM v$active_services order by 1;

NAME                                                                 CON_ID
---------------------------------------------------------------- ----------
SYS$BACKGROUND                                                            1
SYS$USERS                                                                 1
oracle21cdb                                                               1
oracle21cdbXDB                                                            1
oracle21dbcpdb                                                            3
pdb2                                                                      4

6 rows selected.

SQL>
```

- In short, this query gives you the status of all PDBs in the CDB.
- The pdb created is mounted = not yet open for use.

SQL> select name, open_mode from v$pdbs;

- The command lists active services and their container IDs (CDB or PDB), sorted by service name.

SQL> SELECT NAME, CON_ID FROM v$active_services ORDER BY 1;

Display Current Container – CDB$ROOT

Command Prompt - sqlplus s

```
SQL> ALTER PLUGGABLE DATABASE PDB2 OPEN;

Pluggable database altered.

SQL> SHOW CON_NAME;

CON_NAME
------------------------------
CDB$ROOT
SQL>
SQL>
SQL> SELECT name, open_mode from v$pdbs;

NAME
--------------------------------------------------------------------------------
OPEN_MODE
----------
PDB$SEED
READ ONLY

ORACLE21DBCPDB
READ WRITE

PDB2
READ WRITE
```

- This command will return the name of the current container you are connected to, which could be the root container (CDB$ROOT) or a specific pluggable database (PDB).

SQL> Show con_name;
or
SQL> SELECT SYS_CONTEXT('USERENV', 'CON_NAME') FROM DUAL;

SQL*Plus

- Retrieves the names and open modes (like READ WRITE, READ ONLY) of all pluggable databases (PDBs) in the Oracle container.

SQL>SELECT name, open_mode from v$pdbs;

Alter PDB2 – Open & Save

```
SQL> ALTER PLUGGABLE DATABASE PDB2 OPEN;
ALTER PLUGGABLE DATABASE PDB2 OPEN
*
ERROR at line 1:
ORA-65019: pluggable database PDB2 already open


SQL> -- Save the state so PDB2 opens automatically on CDB startup
SQL> ALTER PLUGGABLE DATABASE PDB2 SAVE STATE;

Pluggable database altered.

SQL>
```

SQL*Plus

- After creating the PDB, it will be in a mounted state. The command opens the **PDB2** pluggable database.

SQL> ALTER PLUGGABLE DATABASE PDB2 OPEN;

- The command saves PDB pdb2's state for automatic opening in the same mode (READ ONLY or READ WRITE) on the next CDB startup.

SQL> ALTER PLUGGABLE DATABASE pdb2 SAVE STATE;

Oracle Instance

```
SQL> SELECT instance_name FROM v$instance;

INSTANCE_NAME
----------------
oracle21cdb
```

- This Query retrieves the name of the current Oracle database instance you are connected to.

SQL> SELECT INSTANCE_NAME FROM V$INSTANCE

```
Command Prompt - sqlplus s

Microsoft Windows [Version 10.0.22631.4037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Eric>sqlplus sys@ORACLE21CDB as sysdba

SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 13:29:50 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle.  All rights reserved.

Enter password:

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL>
```

- The command sqlplus sys@ORACLE21CDB as sysdba; connects you to the Oracle database ORACLE21CDB using the SYS user with SYSDBA privileges. This allows you to perform administrative tasks on the container database (CDB) or any pluggable database (PDB) within it.

SQL> Sqlplus sys@ORACLE21CDB as sysdba;

Switch to the PDB & Create User & Grant Privileges

```
Command Prompt - sqlplus s

SQL> SELECT instance_name FROM v$instance;

INSTANCE_NAME
----------------
oracle21cdb

SQL> ALTER SESSION SET CONTAINER = PDB2;

Session altered.

SQL> CREATE USER plsqlauca IDENTIFIED BY password;

User created.

SQL> GRANT ALL PRIVILEGES TO plsqlauca;

Grant succeeded.

SQL>
```

SQL*Plus

- The command switches your current session to the pluggable database (PDB) named PDB2. This allows you to execute commands and queries specifically within that PDB context.

SQL> ALTER SESSION SET CONTAINER = PDB2;

- The command creates a new user named plsqlauca in the Oracle database, with the specified password. After executing this command, you may need to grant the necessary privileges to the user to allow them to perform actions within the database.

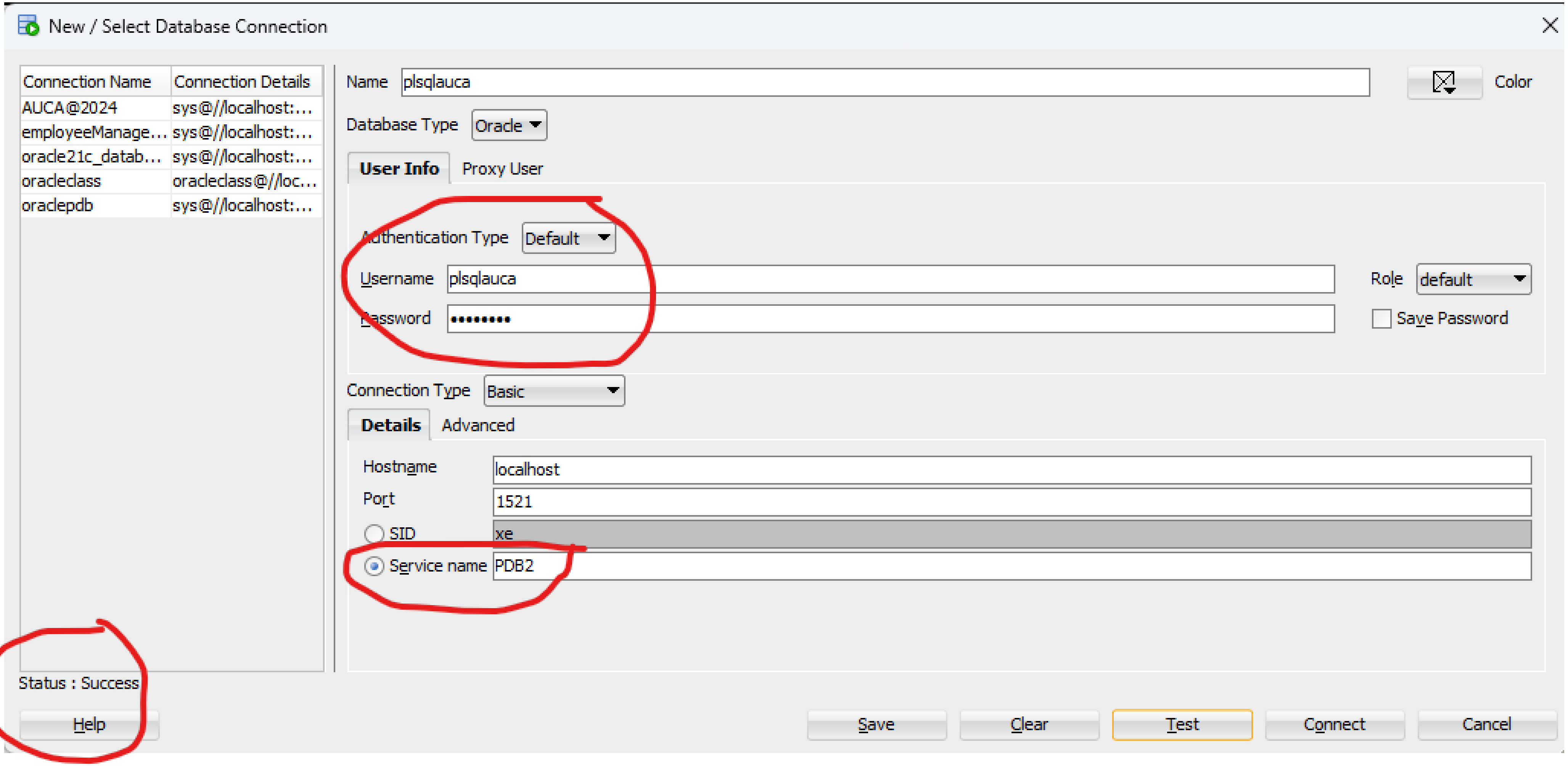
SQL> CREATE USER plsqlauca IDENTIFIED BY password;

- The command GRANT ALL PRIVILEGES TO plsqlauca; grants all available privileges to the user plsqlauca in the Oracle database. This allows the user to perform any action on the database, including creating, modifying, and deleting objects.

SQL> GRANT ALL PRIVILEGES TO plsqlauca

Connecting to Oracle PDB through Oracle SQL Developer for the Newly Created User

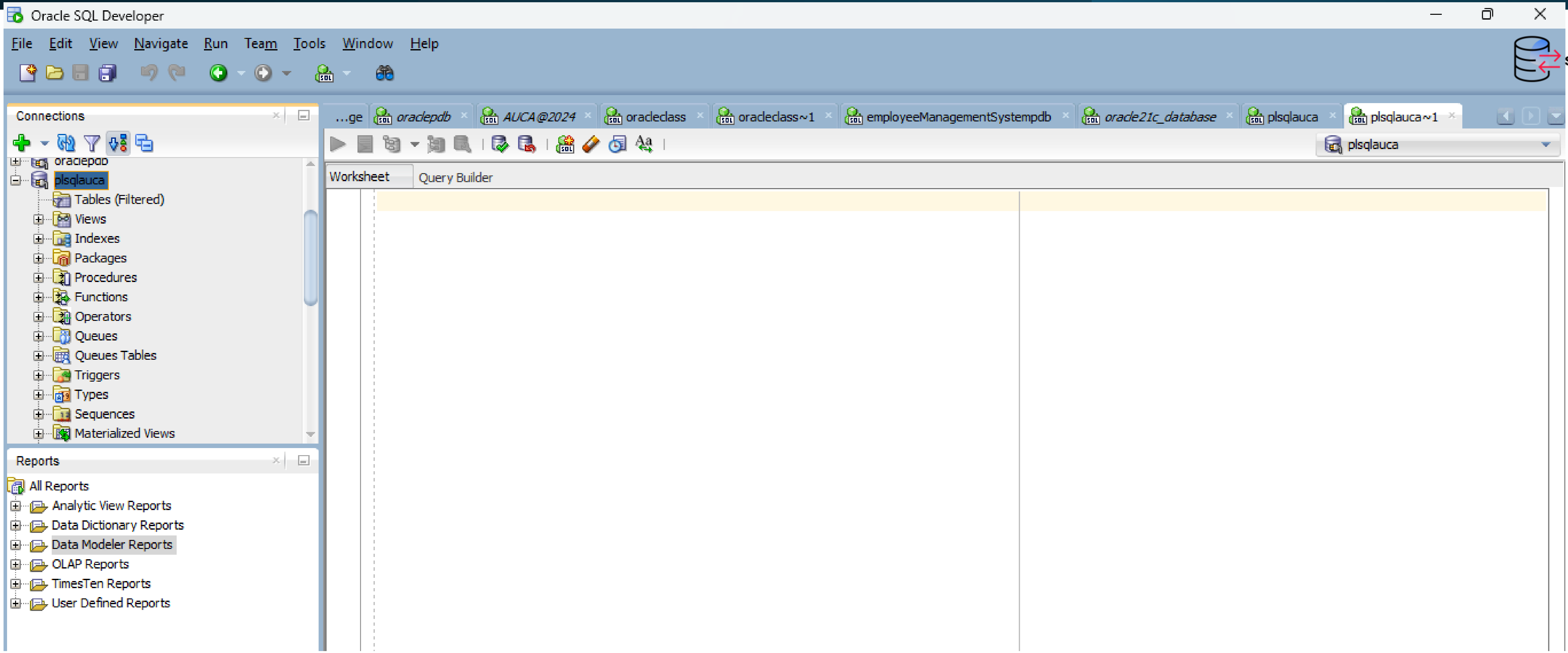
SQL*Plus



Oracle PL/SQL: User Creation and Privilege Grants

Command	Description
CREATE USER XXXXXX IDENTIFIED BY password;	Creates a new user with the given username (XXXXXX) and password.
GRANT ALL PRIVILEGES TO XXXXX	Granting all privileges to a user provides them with full access to the database.
GRANT CONNECT TO XXXXXX; GRANT CREATE SESSION TO XXXXXX;	Grants basic privileges for connecting to the database and creating a session.
GRANT DBA TO XXXXXX; GRANT RESOURCE TO XXXXXX; GRANT UNLIMITED TABLESPACE TO XXXXXX;	Grants DBA privileges, resource management, and unlimited tablespace usage.
GRANT CREATE VIEW TO XXXXXX; GRANT CREATE PROCEDURE TO XXXXXX; GRANT CREATE SEQUENCE TO XXXXXX; GRANT CREATE TRIGGER TO XXXXXX;	Allows the user to create views, procedures, sequences, and triggers.
GRANT ALTER ANY TABLE TO XXXXXX; GRANT ALTER ANY PROCEDURE TO XXXXXX; GRANT ALTER ANY TRIGGER TO XXXXXX;	Grants the ability to alter tables, procedures, and triggers.
GRANT DELETE ANY TABLE TO XXXXXX;	Allows the user to delete rows from any table in the database.
GRANT DROP ANY PROCEDURE TO XXXXXX; GRANT DROP ANY TRIGGER TO XXXXXX; GRANT DROP ANY VIEW TO XXXXXX;	Enables the user to drop (remove) procedures, triggers, and views.

Open worksheet for user: plsqlauca



SQL*Plus

Steps to Permanently Delete Pluggable Databases in Oracle

- **Close the PDBs**: Ensure the pluggable databases are not in use.
- **Unmount the PDBs**: Detach the PDBs from the container database.
- **Use the DROP PLUGGABLE DATABASE Command**: Execute with the INCLUDING DATAFILES option to remove both the PDBs and their associated data files.
- **Permanent Deletion**: This process permanently deletes the databases.

Step 1: Delete the pluggable databases HR and PDB4, follow these steps

Command Prompt - sqlplus s

```
C:\Users\Eric>sqlplus sys/admin@localhost:1521/ORACLE21CDB AS SYSDBA

SQL*Plus: Release 21.0.0.0.0 - Production on Wed Sep 25 15:51:22 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle.  All rights reserved.


Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> col name for a20;
SQL> /
SP2-0103: Nothing in SQL buffer to run.
SQL>
SQL> select name, open_mode from v$containers
  2  /

NAME                 OPEN_MODE
-------------------- ----------
CDB$ROOT             READ WRITE
PDB$SEED             READ ONLY
ORACLE21DBCPDB       READ WRITE
HR                   READ WRITE
PDB4                 READ WRITE

SQL>
```

- To connect to an Oracle database using SQL*Plus as the SYS user with SYSDBA privileges.

C:\Users\Eric>sqlplus sys as SYSDBA
or
C:\Users\Eric> sqlplus sys/admin@localhost:1521/ORACLE21CDB AS SYSDBA

- The query retrieves the names and open modes of all pluggable databases in the current container database.

SQL> col name for a20; -- This is for formatting our output.
SQL> select name, open_mode from v$containers

Step 2: Close and check the paths of Unplug DB

```
SQL> ALTER PLUGGABLE DATABASE HR CLOSE IMMEDIATE;

Pluggable database altered.

SQL> ALTER PLUGGABLE DATABASE PDB4 CLOSE IMMEDIATE;

Pluggable database altered.

SQL>
```

- Before dropping a PDB, it needs to be closed.

SQL> ALTER PLUGGABLE DATABASE HR CLOSE IMMEDIATE;
SQL> ALTER PLUGGABLE DATABASE PDB4 CLOSE IMMEDIATE;

```
SQL> SELECT directory_name, directory_path FROM dba_directories;

DIRECTORY_NAME
--------------------------------------------------------------------------------
DIRECTORY_PATH
--------------------------------------------------------------------------------
DATA_PUMP_DIR
C:\oracle21c_config\admin\oracle21cdb\dpdump/

DBMS_OPTIM_LOGDIR
C:\oracle21c/cfgtoollogs

DBMS_OPTIM_ADMINDIR
C:\oracle21c/rdbms/admin
```

- Find or set the correct directory path for unplugging the PDBs.

SQL> SELECT directory_name, directory_path FROM dba_directories;

Step 3: Use the Path in the UNPLUG Command

```
SQL> ALTER PLUGGABLE DATABASE HR UNPLUG INTO 'C:\oracle21c_config\admin\oracle21cdb\dpdump\hr.xml';

Pluggable database altered.
```

```
SQL> ALTER PLUGGABLE DATABASE PDB4 UNPLUG INTO 'C:\oracle21c_config\admin\oracle21cdb\dpdump\pdb4.xml';

Pluggable database altered.

SQL>
```

SQL*Plus

- Before dropping a PDB, it needs to be closed.

SQL> ALTER PLUGGABLE DATABASE HR UNPLUG INTO 'C:\oracle21c_config\admin\oracle21cdb\dpdump\hr.xml’;
SQL> ALTER PLUGGABLE DATABASE PDB4 UNPLUG INTO 'C:\oracle21c_config\admin\oracle21cdb\dpdump\pdb4.xml';

Step 4: Drop Pluggable Databases 'HR' and 'PDB4' and Verify Their Deletion"

```
SQL> DROP PLUGGABLE DATABASE HR INCLUDING DATAFILES;

Pluggable database dropped.

SQL> DROP PLUGGABLE DATABASE PDB4 INCLUDING DATAFILES;

Pluggable database dropped.

SQL>
SQL>
SQL>
SQL> SELECT name, open_mode FROM v$containers;

NAME                 OPEN_MODE
-------------------- ----------
CDB$ROOT             READ WRITE
PDB$SEED             READ ONLY
ORACLE21DBCPDB       READ WRITE
```

- Before dropping a PDB, it needs to be closed.

SQL> DROP PLUGGABLE DATABASE HR INCLUDING DATAFILES;
SQL> DROP PLUGGABLE DATABASE PDB4 INCLUDING DATAFILES;

- The query shows PDB (All pluggable database) names and their open modes.

SQL> SELECT name, open_mode FROM v$containers;

Notes:

- **CDB$ROOT**: The root container of the multitenant database. It cannot be deleted.
- **PDB$SEED**: A template PDB used for creating new PDBs. It cannot be deleted.
- **ORACLE21DBCPDB**: A user-created PDB that can be deleted, but ensure it's not in use and back up important data before proceeding.

Steps for Cloning a Pluggable Database (PDB)

In Oracle, **cloning** refers to creating an exact copy of a database or its components. This process is commonly used for tasks such as backup, testing, or creating development environments without affecting the production database. Cloning helps maintain data consistency while allowing operations on the copied database.

There are several types of cloning in Oracle:

1. **Database Cloning**: Creating a duplicate of an entire Oracle database. This is useful for testing or performing operations without affecting the original database.

2. **PDB (Pluggable Database) Cloning**: In Oracle Multitenant Architecture, you can clone a PDB within the same container database (CDB) or move it to a different CDB. It can be done as either:
    - **Cold cloning** (when the source PDB is closed)
    - **Hot cloning** (while the source PDB is open)

3. **Schema Cloning**: Copying an individual schema from one database to another. This includes all objects within the schema, such as tables, indexes, views, and triggers.

4. **Tablespace Cloning**: Duplication of a specific tablespace. This might involve creating a read-only copy of a tablespace for reporting purposes.

The cloning process can be done manually using commands like RMAN (Recovery Manager), Data Pump, or through Oracle Enterprise Manager, depending on the specific need.

Step 1: Loging as SYSDBA & Check available PDB

```
Command Prompt - sqlplus s

Microsoft Windows [Version 10.0.22631.4037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Eric>sqlplus sys/admin@localhost:1521/ORACLE21CDB AS SYSDBA

SQL*Plus: Release 21.0.0.0.0 - Production on Wed Sep 25 16:47:45 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle.  All rights reserved.


Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> col name for a20;
SQL> select name, open_mode from v$containers
  2  /

NAME                 OPEN_MODE
-------------------- ----------
CDB$ROOT             READ WRITE
PDB$SEED             READ ONLY
ORACLE21DBCPDB       READ WRITE

SQL>
```

- To connect to an Oracle database using SQL*Plus as the SYS user with SYSDBA privileges.

SQL> sqlplus sys as SYSDBA
or
C:\Users\Eric> sqlplus sys/admin@localhost:1521/ORACLE21CDB AS SYSDBA

- The query retrieves the names and open modes of all pluggable databases in the current container database.

SQL> col name for a20; -- This is for formatting our output.
SQL> select name, open_mode from v$containers

Step 2: Check if You Are on the Root Container and Display All PDBs

```
SQL>
SQL>
SQL> show con_name;

CON_NAME
------------------------------
CDB$ROOT
SQL>
SQL> show pdbs;

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 ORACLE21DBCPDB                 READ WRITE NO
SQL>
```

- This command will return the name of the current container you are connected to, which could be the root container (CDB$ROOT) or a specific pluggable database (PDB).

SQL> Show con_name;
or
SQL> SELECT SYS_CONTEXT('USERENV', 'CON_NAME') FROM DUAL;

- Oracle instance that is currently running, which is usually the same as the **SID** (System Identifier) of the database.

SQL> SHOW PDBS;

Step3: Identify the right path of our PDB

```
SQL>
SQL> SELECT CON_ID, TABLESPACE_NAME, FILE_NAME FROM CDB_DATA_FILES WHERE CON_ID = 3;

    CON_ID TABLESPACE_NAME
---------- ------------------------------
FILE_NAME
--------------------------------------------------------------------------------
         3 SYSTEM
E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\SYSTEM01.DBF

         3 SYSAUX
E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\SYSAUX01.DBF

         3 UNDOTBS1
E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\UNDOTBS01.DBF


    CON_ID TABLESPACE_NAME
---------- ------------------------------
FILE_NAME
--------------------------------------------------------------------------------
         3 USERS
E:\ORACLE21CDATA\ORACLE21CDB\ORACLE21DBCPDB\USERS01.DBF
```

- Retrieves the tablespace names and data file locations for the PDB with CON_ID = 3. Based on the result, the **ORACLE21DBCPDB** (Pluggable Database 1) is located at the following file paths:

SQL> SELECT CON_ID, TABLESPACE_NAME, FILE_NAME FROM CDB_DATA_FILES WHERE CON_ID = 3;

- This path is commonly used for organizing database files, making it easier to manage multiple PDBs within a CDB.

Step 4: Clone the PDB

```
SQL>
SQL> CREATE PLUGGABLE DATABASE plsql2024
  2  FROM ORACLE21DBCPDB
  3  FILE_NAME_CONVERT = ('E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\',
  4                      'E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\plsql2024\');

Pluggable database created.

SQL>
SQL>
SQL>
SQL> show pdbs;

    CON_ID CON_NAME                       OPEN MODE  RESTRICTED
---------- ------------------------------ ---------- ----------
         2 PDB$SEED                       READ ONLY  NO
         3 ORACLE21DBCPDB                 READ WRITE NO
         5 PLSQL2024                      MOUNTED
SQL>
SQL> ALTER PLUGGABLE DATABASE plsql2024 OPEN;

Pluggable database altered.

SQL>
```

- Command to clone existing PDB **“ORACLE21DBCPDB”** to a new one called **plsql2024**

SQL> CREATE PLUGGABLE DATABASE plsql2024
FROM ORACLE21DBCPDB
FILE_NAME_CONVERT =
('E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\',

'E:\oracle21cdata\ORACLE21CDB\oracle21dbcpdb\plsql2024\');

- Check existing PDB, plsql2024 is created with mounted status, which means that not yet opend.
- Open the new PDB (plsql2024)

SQL> show pdbs;
SQL> ALTER PLUGGABLE DATABASE plsql2024 OPEN;

Step 5: Save the State (optional) & Create User in the New PDB

```
SQL> show con_name;

CON_NAME
------------------------------
CDB$ROOT
SQL>
SQL> ALTER SESSION SET CONTAINER=plsql2024;

Session altered.

SQL> CREATE USER adminpdb IDENTIFIED BY admin;

User created.

SQL> GRANT ALL PRIVILEGES TO adminpdb;

Grant succeeded.

SQL>
SQL>
SQL> SELECT username FROM dba_users WHERE username = 'ADMINPDB';

USERNAME
--------------------------------------------------------------------------------
ADMINPDB

SQL>
```

- Save the state: Ensure the PDB opens automatically upon the next startup of the CDB:

SQL> ALTER PLUGGABLE DATABASE plsql2024 SAVE STATE;

- Shift from the root Container to the new DB

SQL> ALTER SESSION SET CONTAINER=plsql2024;

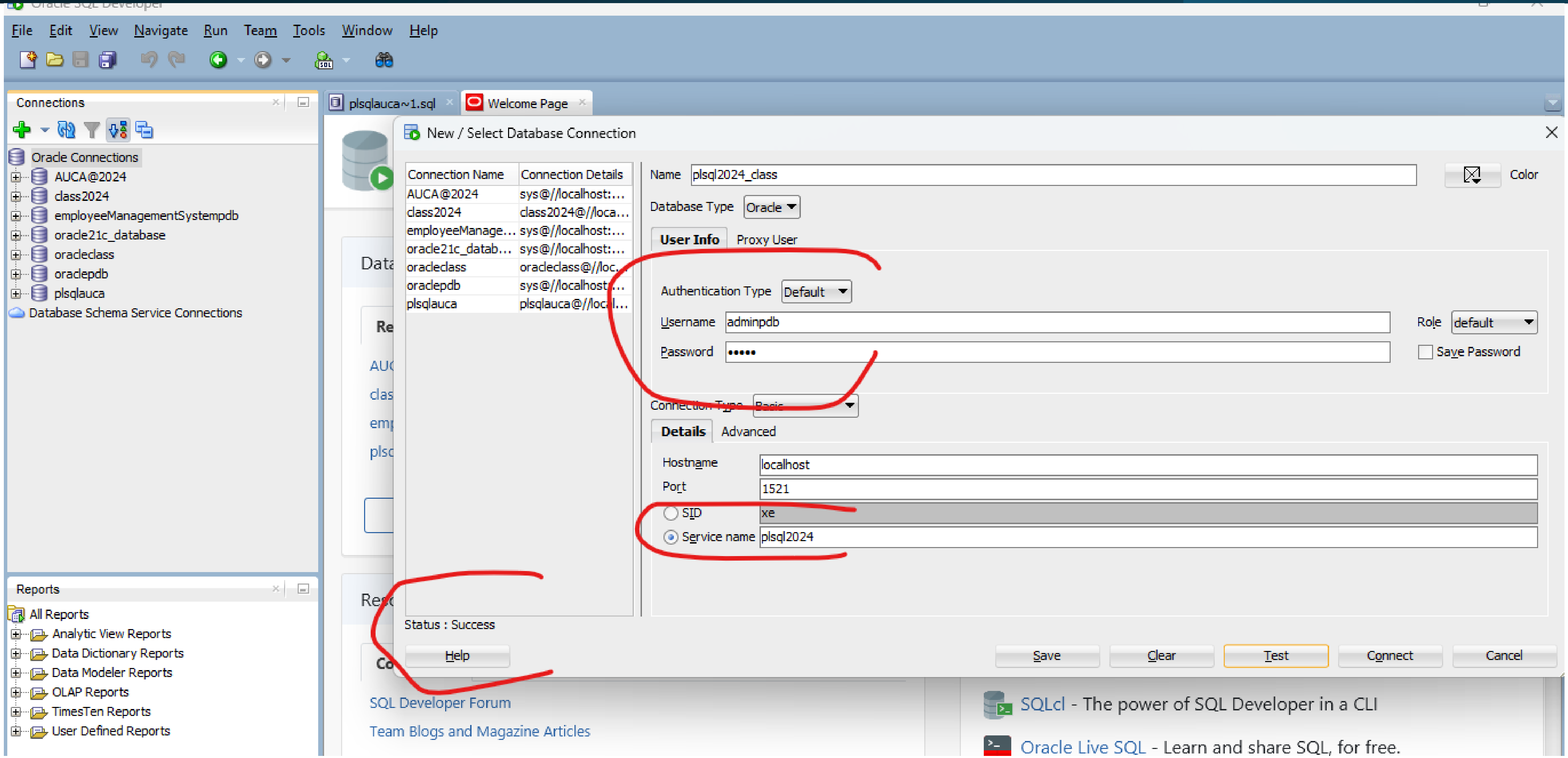
- Create a new user for our new Database plsql2024

SQL> CREATE USER adminpdb IDENTIFIED BY admin;

- Grant all privileges to the new user

SQL> GRANT ALL PRIVILEGES TO admin;
SQL> Display the user created

Step 6: Connect my new user to sql developer



Note on Database User Management

- After configuring the database and creating a user with all privileges, you can interact with the databases effectively.
- Be cautious when granting privileges to users:
    - **Security Risks**: Excessive permissions can lead to unauthorized access.
    - **Data Integrity Issues**: Improper access can compromise data integrity.
- Follow the principle of least privilege:
    - Grant only the permissions necessary for users to perform their tasks.
- Regularly review user privileges:
    - Adjust permissions as needed to maintain a secure database environment.

Oracle Enterprise Manager

SQL*Plus

Oracle Enterprise Manager (OEM) is a comprehensive management tool **designed for monitoring, managing, and optimizing Oracle environments, including databases, middleware, and applications**. Here are some key features:

Monitoring

- OEM provides real-time monitoring of database performance, availability, and resource utilization.

Performance Tuning

- It enables administrators to perform tasks such as backup and recovery, patch management, and configuration management.

Reporting

- It includes reporting features to generate insights on performance metrics, user activities, and system health.

Centralized Control

- It provides a centralized interface to manage multiple Oracle instances across different environments, facilitating easier administration.

Overall, Oracle Enterprise Manager is essential for ensuring the health and efficiency of Oracle database environments.

How to open Oracle Enterprise Manager: Connect to Sys

Windows PowerShell

```
PS C:\Users\Eric> sqlplus sys as sysdba

SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 22:32:16 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle.  All rights reserved.

Enter password:

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL> show con_name;

CON_NAME
------------------------------
CDB$ROOT
SQL> ALTER SESSION SET CONTAINER = CDB$ROOT;

Session altered.

SQL> SELECT SYS_CONTEXT('USERENV', 'CON_NAME') AS current_container FROM dual;

CURRENT_CONTAINER
--------------------------------------------------------------------------------
CDB$ROOT
```

SQL*Plus

In this:

- **Connect to the System**

SQL> sqlplus sys as sysdba

- **Check if I am connected to the root container**

SQL> show con_name

- If not, I should use:

SQL> ALTER SESSION SET CONTAINER = CDB$ROOT;
Or

- Check the current container

SQL> SELECT SYS_CONTEXT('USERENV', 'CON_NAME') AS current_container FROM dual;

How to open Oracle Enterprise Manager: Verify configuration & set HTTP(s)

```
SQL> SELECT DBMS_XDB_CONFIG.GETHTTPPORT() AS HTTP_PORT,
  2         DBMS_XDB_CONFIG.GETHTTPSPORT() AS HTTPS_PORT
  3  FROM dual;

 HTTP_PORT HTTPS_PORT
---------- ----------
         0          0

SQL>
SQL> BEGIN
  2      DBMS_XDB_CONFIG.SETHTTPPORT(8080);
  3      DBMS_XDB_CONFIG.SETHTTPSPORT(8443);  -- Optional for HTTPS
  4  END;
  5  /

PL/SQL procedure successfully completed.

SQL>
SQL>
SQL> SELECT DBMS_XDB_CONFIG.GETHTTPPORT() AS HTTP_PORT,
  2         DBMS_XDB_CONFIG.GETHTTPSPORT() AS HTTPS_PORT
  3  FROM dual;

 HTTP_PORT HTTPS_PORT
---------- ----------
      8080       8443
```

- **Verify Configuration: After restarting, run the initial query again to confirm the new port settings.**
- SELECT DBMS_XDB_CONFIG.GETHTTPPORT() AS HTTP_PORT, DBMS_XDB_CONFIG.GETHTTPSPORT() AS HTTPS_PORT FROM dual;

- Since both HTTP_PORT and HTTPS_PORT are returning 0, it confirms that the HTTP(S) server is currently not enabled. Here's how to proceed:
- Set New HTTP(S) Ports: Let's set the HTTP and HTTPS ports to a specific value. Choose ports that are not in use (for example, 8080 for HTTP and 8443 for HTTPS):

```
SQL> BEGIN
    DBMS_XDB_CONFIG.SETHTTPPORT(8080);
    DBMS_XDB_CONFIG.SETHTTPSPORT(8443);  -- Optional for HTTPS
END;
/
```

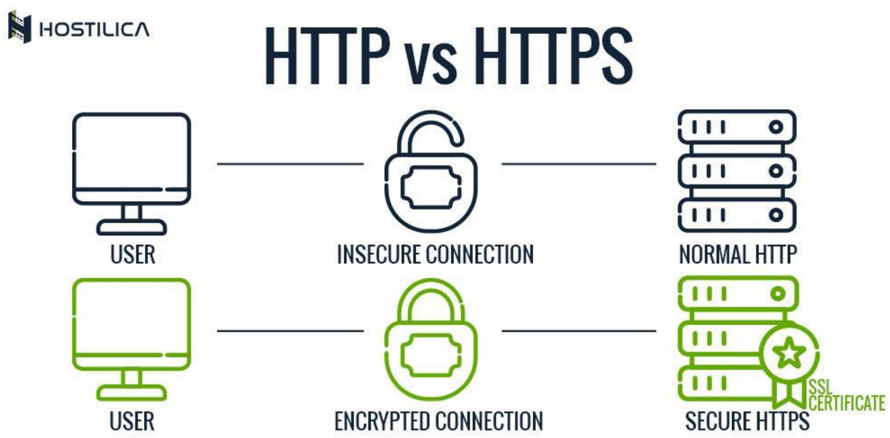
- Check for Errors: If you encounter a port conflict again, try different port numbers.
- Verify Configuration: After executing the above, check the port settings again:

```
SQL> SELECT DBMS_XDB_CONFIG.GETHTTPPORT() AS HTTP_PORT,
       DBMS_XDB_CONFIG.GETHTTPSPORT() AS HTTPS_PORT
FROM dual;
```

Using LSNRCTL to Check Oracle Listener Status

Administrator: Command Prompt

```
C:\Windows\System32>lsnrctl status | findstr HOST
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=Eric-Admin)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=Eric-Admin)(PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=Eric-Admin)(PORT=5500))(Presentation=HTTP)(Session=RAW))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcps)(HOST=Eric-Admin)(PORT=5501))(Security=(my_wallet_directory=C:\ORACLE21C_CONFIG\admin\oracle21cdb\xdb_wallet))(Presentation=HTTP)
(Session=RAW))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=Eric-Admin)(PORT=8080))(Presentation=HTTP)(Session=RAW))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcps)(HOST=Eric-Admin)(PORT=8443))(Security=(my_wallet_directory=C:\ORACLE21C_CONFIG\admin\oracle21cdb\xdb_wallet))(Presentation=HTTP)
(Session=RAW))

C:\Windows\System32>
```



In this:
- **Identify the hostname that correspond to the port number 8443**

C:\Windows\System32>lsnrctl status | findstr HOST

Source Image: https://hostilica.com/blog/http-vs-https/

How to open Oracle Enterprise Manager: Restart database

```
SQL> SHUTDOWN IMMEDIATE;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> STARTUP;
ORACLE instance started.

Total System Global Area 2147482528 bytes
Fixed Size                  9857952 bytes
Variable Size            1392508928 bytes
Database Buffers          738197504 bytes
Redo Buffers                6918144 bytes
Database mounted.
Database opened.
SQL>
SQL> select dbms_xdb_config.gethttpsport() from dual;

DBMS_XDB_CONFIG.GETHTTPSPORT()
------------------------------
                          8443

SQL>
```

- Restart the Database: If the ports were successfully set, restart the database to apply changes:

SQL> SHUTDOWN IMMEDIATE;
STARTUP;

- This will display both port numbers in one result set.

SQL> select dbms_xdb_config.gethttpsport() from dual;

Connecting to Oracle PDB Locally Using SQL*Plus for the Newly Created User

Windows PowerShell

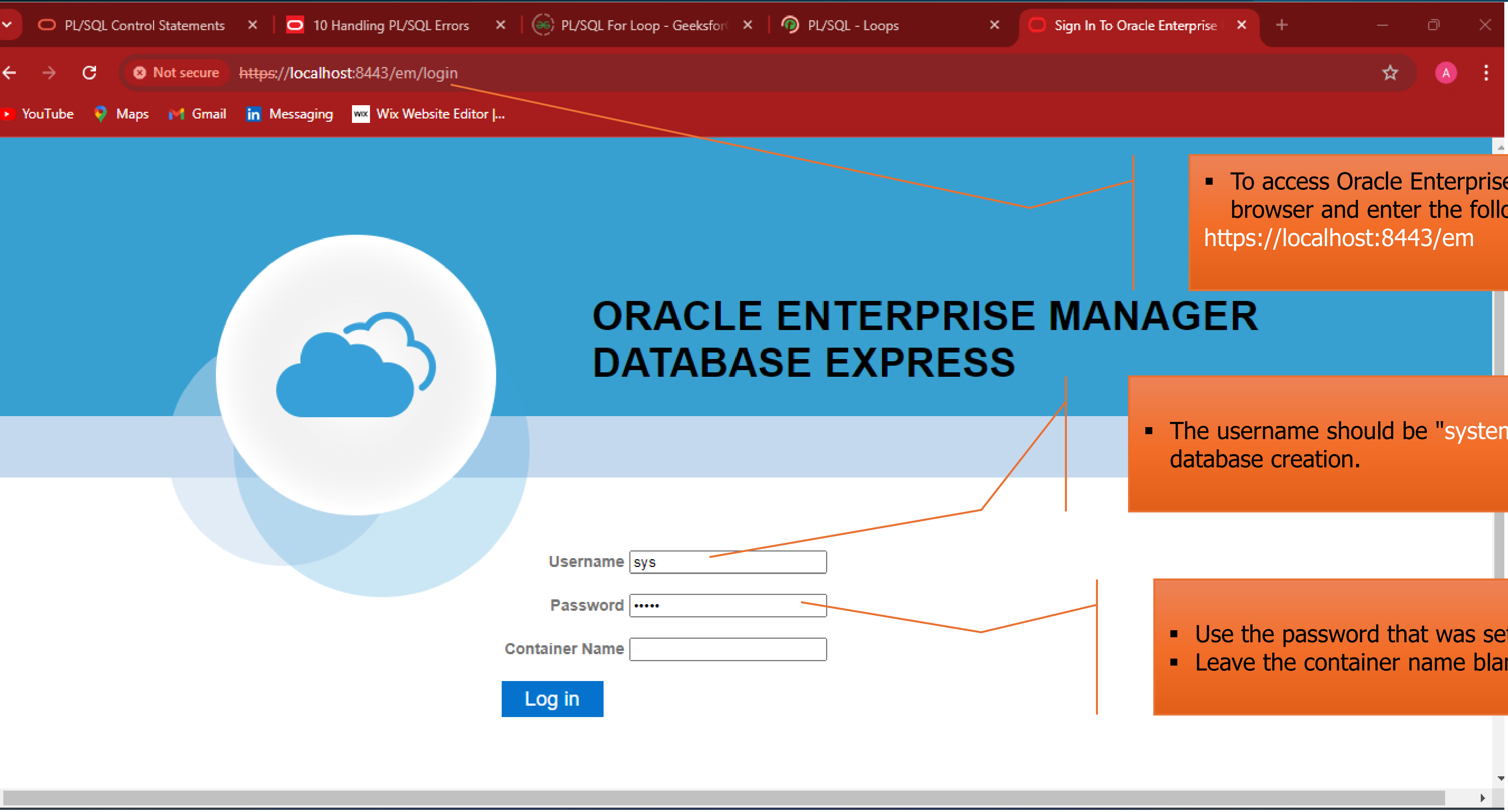
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Eric> sqlplus plsqlauca/password@//localhost:1521/PDB2

SQL*Plus: Release 21.0.0.0.0 - Production on Sun Sep 22 21:09:27 2024
Version 21.3.0.0.0

Copyright (c) 1982, 2021, Oracle.  All rights reserved.

Last Successful login time: Sun Sep 22 2024 19:15:47 +02:00

Connected to:
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0

SQL>
```

In this:

- **plsqlauca:** Username.
- **password:** Password for the plsqlauca user.
- **localhost:** Refers to your local machine.
- **1521:** The default Oracle listener port.
- **PDB2**: The service name of the pluggable database you're connecting to.

SQL> sqlplus plsqlauca/password@//localhost:1521/PDB2

How to open Oracle Enterprise Manager: Restart database



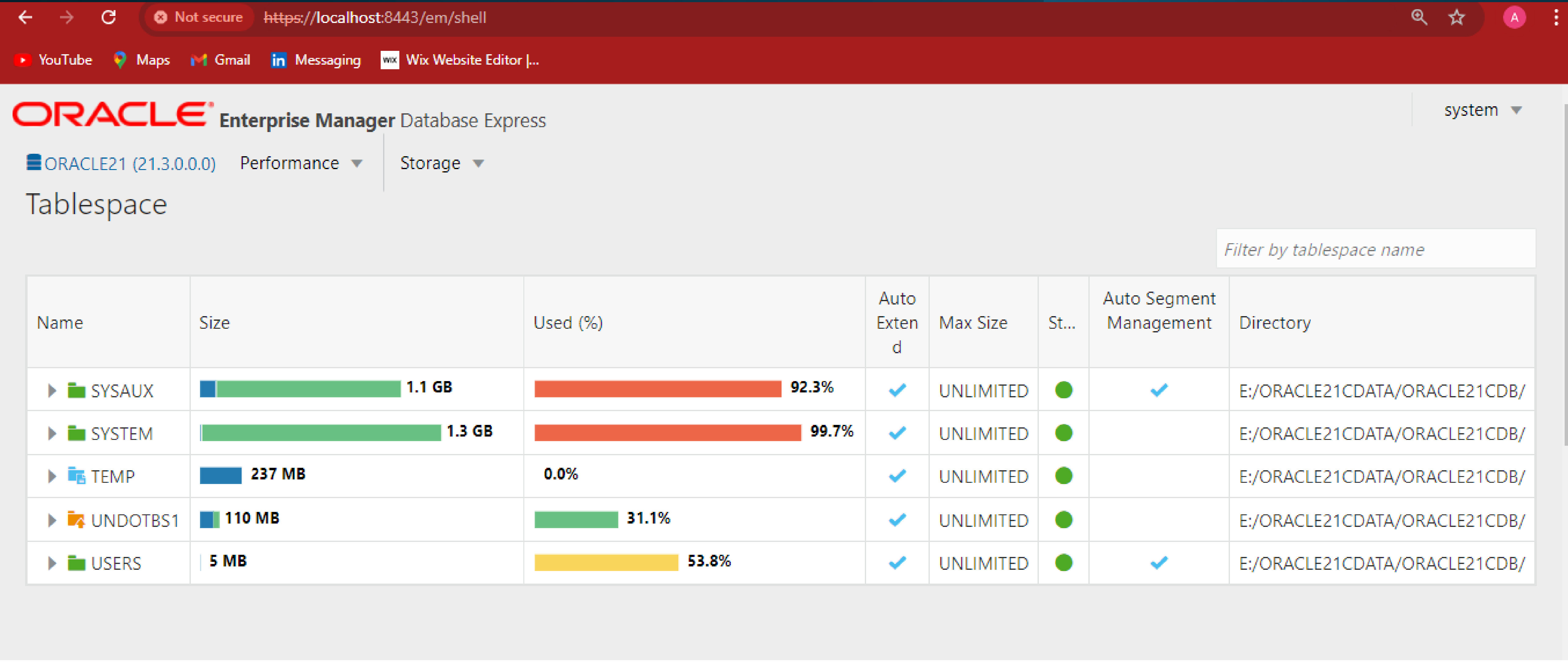
- To access Oracle Enterprise Manager, open your web browser and enter the following URL:
https://localhost:8443/em

- The username should be "system" or "sys" as used during database creation.

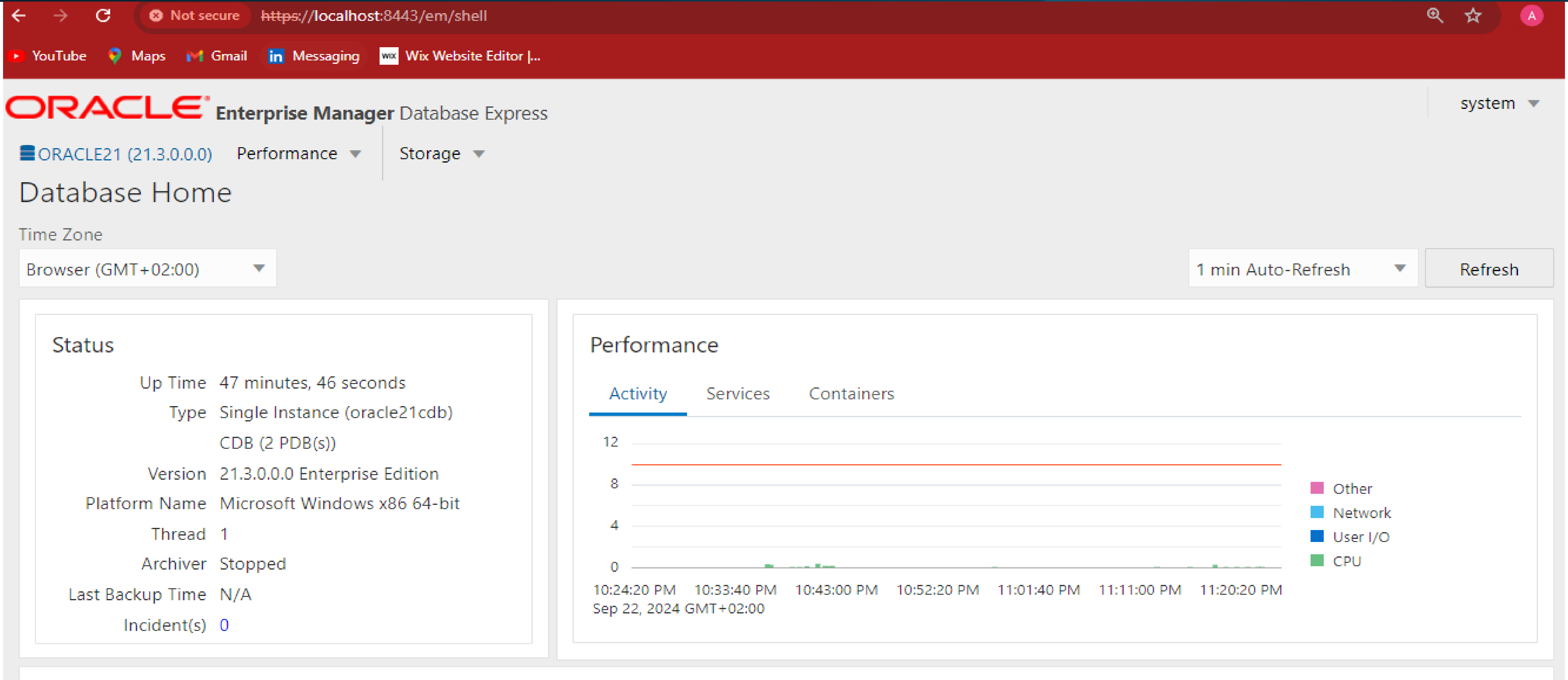
- Use the password that was set during Oracle installation.
- Leave the container name blank.

SQL*Plus

Oracle Enterprise Manager – Tablespaces



Oracle Enterprise Manager – Database Home



It's time for you to explore Oracle Enterprise Manager (OEM) and get hands-on experience!

SQL*Plus

Simplified Administration: OEM offers a unified platform that makes database management intuitive and efficient.

Powerful Tools: You'll have access to robust features for monitoring, managing, and optimizing Oracle environments.

High Availability: OEM helps ensure that databases are always running at peak performance.

Streamlined Operations: Routine tasks are automated, reducing complexity and making your job easier.

Maximized Efficiency: Learning to use OEM will help you get the most out of any Oracle infrastructure.

Assignment II : Assignment: Database Creation, Deletion & EOM (1/2)

SQL*Plus

Task 1: Create a New Pluggable Database (PDB)
Each student is required to create a PDB named plsql_class2024db.

- **Username Format:** Your username should begin with the first two letters of your first name followed by _plsqlauca.
  Example: If your first name is Eric, your username will be er_plsqlauca.
- **Password:** Choose a personal password, preferably something simple for ease of use.
- **Purpose:** This username will be used for class activities where you will store all your work during the course.

Task 2: Create and Delete a Pluggable Database

Instructions:

- You are required to create another Pluggable Database (PDB) based on the first two letters of your first name followed by _to_delete_pdb.
- After creating the PDB, you must delete it.

Example: If your first name is Eric, you will:

- Create a PDB named er_to_delete_pdb.
- Once the PDB is successfully created, proceed to delete it.

Ensure you take screenshots of both the creation and deletion process, and include all SQL commands used.

Task 3: Configure Oracle Enterprise Manager
Configure your Oracle Enterprise Manager and provide a screenshot of the dashboard after completing the above tasks.

Assignment: Database Creation, Deletion & EOM (2/2)

Task 4: Classroom Submission

- Create a GitHub repository for this assignment with a name related to the task.
- Ensure you take screenshots of each step (creating, deleting the PDB, and Oracle Enterprise Manager dashboard).
- Include all the queries you used in the README file of your repository, along with explanations.
- Add me as a collaborator on the repository.
- Note: Use the GitHub username **ericmaniraguha** or the email **ericmaniraguha@gmail.com** when adding me as a collaborator.
- Submit the link to your GitHub repository by filling out this [Google Form].

Grading and Deadlines

- This assignment is worth **10 points**.
- Late submissions will result in a **3-point deduction**.

Start Date: 26th September 2024
Due Date: 3rd October 2024, 11h59 pm

No excuses! Make sure to complete all tasks and submit on time.

Keep Learning!